

Curcumin, an active component of turmeric in the prevention and treatment of ulcerative colitis: preclinical and clinical observations.

Inflammatory bowel disease (IBD) comprising of ulcerative colitis (UC) and Crohn's disease (CD) is a major ailment affecting the small and large bowel. In clinics, IBD is treated using 5-aminosalicylates, antibiotics, the steroids and immunomodulators. Unfortunately, the long term usages of these agents are associated with undue side effects and compromise the therapeutic advantage. Accordingly, there is a need for novel agents that are effective, acceptable and non toxic to humans. Preclinical studies in experimental animals have shown that curcumin, an active principle of the Indian spice turmeric (*Curcuma longa* Linn) is effective in preventing or ameliorating UC and inflammation. Over the last few decades there has been increasing interest in the possible role of curcumin in IBD and several studies with various experimental models of IBD have shown it to be effective in mediating the inhibitory effects by scavenging free radicals, increasing antioxidants, influencing multiple signaling pathways, especially the kinases (MAPK, ERK), inhibiting myeloperoxidase, COX-1, COX-2, LOX, TNF- α , IFN- γ , iNOS; inhibiting the transcription factor NF- κ B. Clinical studies have also shown that co-administration of curcumin with conventional drugs was effective, to be well-tolerated and treated as a safe medication for maintaining remission, to prevent relapse and improve clinical activity index. Large randomized controlled clinical investigations are required to fully understand the potential of oral curcumin for treating IBD.